Lab 4 - Polylines, Arcs, Ellipses and Splines

Course: AutoCAD Masterclass (C391) | Topic 02: Technical Drawing with AutoCAD

Objective: LO2 - produce 2D technical drawings using AutoCAD

Goal

Draw practice drawing EX-3 combining polylines with arc segments, standalone arcs, an ellipse and a spline, plus construction lines to project feature positions between views.

You'll produce

The completed EX-3 drawing with mixed line/arc geometry drawn as clean, connected objects.

Commands / tools: PLINE (Arc mode), ARC, ELLIPSE, SPLINE, XLINE, POINT/DIVIDE

Files in this lab

- ex3.dwg - AutoCAD drawing file
- ex3-drawing-sheet.pdf - drawing sheet / reference (PDF)
- ex3.png - reference image

Step-by-step

1. Open the EX-3 sheet (ex3-drawing-sheet.pdf) and plan the outline as one polyline
2. Draw the outline with PLINE, switching between Line and Arc segments as you go

```
PL → pick → A (arc) → L (line)
```

3. Place construction lines to project the centres of features between views

```
XL → H / V through centres
```

4. Add a three-point arc and a centre-start-end arc where the sheet shows open curves

```
A → 3 points  |  A → C
```

5. Draw the ellipse from its axis endpoints and a smooth spline through the fit points

```
EL → axis ends → half-width  |  SPL
```

6. Compare with ex3.dwg, purge the construction lines and save

```
ERASE xlines → SAVEAS ex3-mine.dwg
```

Check your work

The outline is a single polyline (one object when selected), arcs and the ellipse match the sheet dimensions, and no construction lines remain in the final drawing.